Chapter 6
Eigenvalues and Eigenvectors

6.1 Eigenvalues and Eigenvectors

Motivation

- For the matrix $A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}$, how to calculate $A^{10}\begin{bmatrix} 1 \\ 4 \end{bmatrix}$?
- Note that for $\mathbf{x}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, we have $A\mathbf{x}_1 = \mathbf{x}_1$ and $A\mathbf{x}_2 = \frac{1}{2}\mathbf{x}_2$.
- Then, $A^{10}\mathbf{x}_1 = \mathbf{x}_1$ and $A^{10}\mathbf{x}_2 = \left(\frac{1}{2}\right)^{10}\mathbf{x}_2$
- Since $\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \mathbf{x}_1 + \mathbf{x}_2$,

$$A^{10}\begin{bmatrix} 1 \\ 4 \end{bmatrix} = A^{10}\mathbf{x}_1 + A^{10}\mathbf{x}_2 = \mathbf{x}_1 + \left(\frac{1}{2}\right)^{10}\mathbf{x}_2 \approx \mathbf{x}_1$$

- If we can find special λ and $\mathbf{x}$ so that $A\mathbf{x} = \lambda\mathbf{x}$, then on the line $\mathrm{Span}(\mathbf{x})$, the matrix A acts just like a scalar λ.

Eigenvalues and Eigenvectors

Definition

Let A be an $n \times n$ matrix.

- A scalar λ is said to be an **eigenvalue 特征值** of A if there exists a nonzero vector $\mathbf{x} \neq 0$ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

- Such $\mathbf{x} \neq 0$ is said to be an **eigenvector 特征向量** for A belonging to λ.

- The pair $(\lambda, \mathbf{x})$ is sometimes called an eigenpair for A.
- Eigenvalues and eigenvectors are only defined for square matrices
- Eigenvectors **must be nonzero**

Example: Eigenvalues and Eigenvectors

$$A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

1. Show that $\mathbf{x}$ is an eigenvector of A. What eigenvalue does it belong to?
2. Is $2\mathbf{x} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ an eigenvector of A? What eigenvalue does it belong to?

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Solution:

1. $\begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, so $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$, belonging to eigenvalue $\lambda = 3$.
2. $\begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 6 \end{bmatrix} = 3\begin{bmatrix} 4 \\ 2 \end{bmatrix}$, so $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$ is also an eigenvector, belong to the same eigenvalue $\lambda = 3$.

Some Properties (as exercises)

Let λ be an eigenvalue of A and let $\mathbf{x} \neq \mathbf{0}$ be an eigenvector belonging to λ, i.e. $A\mathbf{x} = \lambda\mathbf{x}$. Then,

Think about why!

- [Ex 3] A is singular $\Leftrightarrow \lambda = 0$ is an eigenvalue of A.
- [Ex 4] If A is nonsingular, then λ^{-1} is an eigenvalue of A^{-1}, and $\mathbf{x}$ is an eigenvector of A^{-1} belonging to λ^{-1}, i.e. $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.
- [Ex 6] For $m \geq 1$, λ^m is an eigenvalue of A^m, and $\mathbf{x}$ is an eigenvector of A^m belonging to λ^m, i.e. $A^m\mathbf{x} = \lambda^m\mathbf{x}$.
- [Ex 8] If $A^2 = A$, what can λ be? Hint: $A^2\mathbf{x} = A\mathbf{x}$.
- [Ex 9] If $A^{30} = O$, what can λ be? Hint: $A^{30}\mathbf{x} = O\mathbf{x}$

Finding eigenvalues and eigenvectors

- Note that

$$A\mathbf{x} = \lambda\mathbf{x} \Leftrightarrow A\mathbf{x} - \lambda\mathbf{x} = 0 \Leftrightarrow (A - \lambda I)\mathbf{x} = 0 \Leftrightarrow \mathbf{x} \in N(A - \lambda I)$$

- $N(A - \lambda I)$: **eigenspace 特征空间** corresponding to the eigenvalue λ.

This has a nontrivial solution $\mathbf{x} \neq \mathbf{0}$ if and only if we pick λ so that the square matrix $(A - \lambda I)$ is singular.

That is: $\det(A - \lambda I) = 0$.

Properties of Eigenvalue

Let A be an $n \times n$ matrix and λ be a scalar.

The following statements are equivalent:

(a) λ is an eigenvalue of A.

(b) $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has nontrivial solutions. Exists nonzero eigenvector

(c) $N(A - \lambda I) \neq \{\mathbf{0}\}$ Nontrivial eigenspace

(d) $A - \lambda I$ is singular.

(e) $\det(A - \lambda I) = 0$ **Characteristic equation**

- $p(\lambda) = \det(A - \lambda I)$ is a degree n polynomial in variable λ, called the **characteristic polynomial 特征多项式**.
- $\det(A - \lambda I) = 0$: **characteristic equation 特征方程** for A.
- First use $\det(A - \lambda I) = 0$ to find all eigenvalues for A, and then use $N(A - \lambda I)$ to find all eigenvectors for A.

Example: Finding eigenvalues

1. Find all eigenvalues for $A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$.
2. Find the eigenvectors corresponding to each eigenvalue.

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Solution:

1. Eigenvalues λ satisfy the characteristic equation:

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 2 \\ 3 & -2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)(-2-\lambda) - 6 = \lambda^2 - \lambda - 12 = 0$$

So, the eigenvalues of A are $\lambda_1 = 4, \lambda_2 = -3$.

Example: Finding eigenvectors

1. Find all eigenvalues for $A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$. $\lambda_1 = 4, \lambda_2 = -3$
2. Find the eigenvectors corresponding to each eigenvalue.

Solution (cont.):

2. To find the eigenvectors, we consider the null spaces $N(A - \lambda I)$.

$$\lambda_1 = 4, A - 4I = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$$

Solving $(A - 4I)\mathbf{x} = \mathbf{0}$, we find that $x_1 = 2x_2$, that is

$$N(A - 4I) = \left\{ \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} \right\} = \text{Span}\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} \right)$$

So, $\{[2,1]^T\}$ is a basis for the eigenspace corresponding to $\lambda_1 = 4$.

Similarly, the eigenspace corresponding to $\lambda_2 = -3$ is

$$N(A + 3I) = N\left(\begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix} \right) = \text{Span}\left(\begin{bmatrix} -1 \\ 3 \end{bmatrix} \right)$$

Example: 3×3 matrix eigenspaces

Find the eigenvalues and corresponding eigenspaces for

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$$

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Solution:

- Characteristic polynomial:

$$p(\lambda) = \begin{vmatrix} 3-\lambda & -1 & -2 \\ 2 & -\lambda & -2 \\ 2 & -1 & -1-\lambda \end{vmatrix}$$

$$= (3-\lambda)(-\lambda)(-1-\lambda) + (-1)(-2)(2) + (-2)(2)(-1)$$
$$-(2)(\lambda)(2) - (-1)(-2)(3-\lambda) - (-1-\lambda)(2)(-1)$$
$$= -\lambda^3 + 2\lambda^2 - \lambda = -\lambda(\lambda-1)^2$$

- Eigenvalues: $p(\lambda) = 0 \Rightarrow \lambda_1 = 0, \lambda_2 = \lambda_3 = 1$

Example: 3×3 matrix eigenspaces

Find the eigenvalues and corresponding eigenspaces for $\lambda_1 = 0, \lambda_2 = \lambda_3 = 1$

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$$

Solution (cont):

- Eigenspace for $\lambda_1 = 0$ is just $N(A)$:

$$\left[\begin{array}{ccc|c} 3 & -1 & -2 & 0 \\ 2 & 0 & -2 & 0 \\ 2 & -1 & -1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 1 & 0 & -1 & 0 \\ 2 & -1 & -1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\Rightarrow x_1 - x_3 = 0, x_2 - x_3 = 0 \Rightarrow x_1 = x_2 = x_3 \Rightarrow \mathbf{x} = [a, a, a]^T$

Therefore, the eigenspace corresponding to $\lambda_1 = 0$ is

$$\text{Span}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right).$$

Example: 3 × 3 matrix eigenspaces

Find the eigenvalues and corresponding eigenspaces for $\lambda_1 = 0, \lambda_2 = \lambda_3 = 1$

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$$

Solution (cont):

- Eigenspace for $\lambda = 1$ is $N(A - I)$:

$$\left[\begin{array}{ccc|c} 3-1 & -1 & -2 & 0 \\ 2 & -1 & -2 & 0 \\ 2 & -1 & -1-1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 2 & -1 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & -1/2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow x_1 = \left(\frac{1}{2}\right)x_2 + x_3 \Rightarrow \mathbf{x} = \begin{bmatrix} \frac{\alpha}{2} + \beta \\ \alpha \\ \beta \end{bmatrix} = \frac{\alpha}{2}\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Questions: preview for 6.3

- If we have different eigenvalues $\lambda_1, \lambda_2, \lambda_3$, are their corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ linearly independent?
- Can we find a basis of $\mathbb{R}^n$ consisting of only eigenvectors of A?
  (If we can, then A becomes very simple under this basis!)
- Can we put all n eigenvalues λ_i and n corresponding eigenvectors $\mathbf{x}_i$ into one short matrix equation?
- How to calculate matrix powers A^m easily?